

nehaber.dev: A BERT-Based Feature-Level Hybrid Platform for Turkish Fake News Detection

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Abstract—This study presents *nehaber.dev*, an integrated platform for automated Turkish fake-news analysis. The system employs a two-stage architecture. In Stage 1, incoming text is represented using a 768-dimensional Turkish Sentence-BERT embedding and compared with labeled records through pgvector cosine-distance search. If no sufficiently similar record is found, Stage 2 concatenates the embedding with eight handcrafted natural language processing signals to form a 776-dimensional feature vector and applies a StandardScaler–Logistic Regression pipeline. In the current Stage 2 local policy, classifier probability determines the binary local result, while the rule-based risk score is stored separately for interpretation and monitoring. Gemini 2.5 Flash then performs an asynchronous evidence-backed review. Its prompt mode depends on the local confidence score, but a differing Gemini verdict can update the stored result in either mode. On a fixed dataset of 3,764 samples, five-fold stratified cross-validation yielded an accuracy of 0.8727 ± 0.0065 and a Macro-F1 score of 0.8724 ± 0.0065 for the local feature-level classifier. The eight signals produced a numerical Macro-F1 increase of 0.0061 over BERT-only features, whereas TF-IDF with Logistic Regression obtained 0.8745 ± 0.0095 . These metrics correspond to the local Stage 2 classifier. The platform also integrates Stage 1 semantic retrieval, Gemini-assisted evidence review, URL and uploaded-image analysis, asynchronous processing, RSS-based ingestion, and community verification.

Index Terms—fake news detection, Turkish natural language processing, BERT, feature-level fusion, vector similarity search, community verification

I. INTRODUCTION

The widespread adoption of the internet and smart mobile devices has substantially increased the speed of access to information. However, content can spread much faster than it can be verified. During the COVID-19 pandemic, the World Health Organization described this phenomenon as an *infodemic* and emphasized that health-related misinformation could have consequences comparable to those of the disease itself [1].

In Türkiye, intensive content flows through platforms such as WhatsApp, X, and online news portals create favorable conditions for the rapid spread of disinformation. Deficiencies in digital media literacy and algorithmic filter bubbles can make it more difficult for users to critically evaluate the credibility of encountered content [2]. During elections, natural disasters, pandemics, and other crises, false or misleading news may intensify public anxiety, deepen social polarization, and reduce

trust in public institutions. Human fact-checking organizations such as *teyit.org* and *Doğruluk Payı* provide valuable verification services, but manual workflows have limited scalability. This limitation has motivated both academic and industrial research on automated fake-news detection [3].

Turkish is an agglutinative language in which stems can take numerous suffixes to express grammatical and contextual information. This structure increases the difficulty of tasks such as stemming, morphological analysis, and word-sense disambiguation [4]. Consequently, models and preprocessing strategies designed primarily for English cannot always be transferred directly to Turkish without adaptation.

The term *fake news* is multidimensional and contested. Within the information-disorder framework, problematic information is commonly discussed as misinformation, disinformation, or malinformation according to factual accuracy and intent [5]. The same framework distinguishes satire or parody, misleading content, imposter content, fabricated content, false context, manipulated content, and false connection. For the operational classification task in this study, the output space is simplified to two labels: FAKE and AUTHENTIC. Community-voting labels are maintained as a separate taxonomy and are not treated as direct equivalents of the machine-learning labels.

Early automated approaches used Term Frequency–Inverse Document Frequency (TF-IDF) representations with Support Vector Machines, Naïve Bayes, or Logistic Regression [6]. The LIAR benchmark subsequently enabled fine-grained evaluation using six truthfulness labels [7]. BERT introduced bidirectional contextual representations that substantially improved many natural language processing tasks [8]. Sentence-BERT later enabled efficient sentence-level semantic similarity computation [9].

The model used in this study, *emrecaan/bert-base-turkish-cased-mean-nli-stsb-tr*, follows Sentence-BERT principles and was fine-tuned on Turkish Natural Language Inference and Semantic Textual Similarity tasks [10]. It is not used as a directly fine-tuned fake-news classifier. Instead, it generates 768-dimensional sentence embeddings that support both semantic retrieval and downstream Logistic Regression classification.

Hybrid fake-news detection approaches often combine textual representations with additional contextual or handcrafted

features [11]. In this study, the term *hybrid* refers primarily to feature-level fusion: a 768-dimensional semantic representation is concatenated with eight handcrafted linguistic signals. The resulting 776-dimensional vector is classified using Logistic Regression. Dense-vector systems also require efficient nearest-neighbor search [12]. pgvector was selected because it supports relational data and vector search within PostgreSQL and provides IVFFlat and HNSW indexing options [13].

To the best of our knowledge, based on the academic studies and publicly available systems reviewed in this work, few open-source platforms integrate Turkish-language semantic retrieval, automated classification, LLM-assisted evidence generation, and community verification within a single operational system. nehaber.dev was developed to investigate this integrated design.

II. MATERIALS AND METHODS

A. System Overview

The platform combines automatic embedding generation, semantic retrieval, feature-level classification, asynchronous background processing, URL and uploaded-image analysis, RSS ingestion, and community verification. Its principal stakeholders are end users who submit content for analysis, moderators who manage community and security processes, and autonomous services responsible for scheduled ingestion, model execution, and maintenance tasks.

B. Stakeholder Analysis

a) *End User*: The end user submits text, a Uniform Resource Locator (URL), or an uploaded image and expects a concise result, supporting evidence, and access to prior analyses and community discussions.

b) *Editor or Moderator*: Moderators require tools for user management, content moderation, security-event review, and system monitoring.

c) *Autonomous Services*: Background services collect RSS content, generate embeddings, execute analyses, update categories, schedule maintenance tasks, and process user feedback for model retraining.

C. System Requirements

The main functional and non-functional requirements are summarized in Tables I and II.

D. Representative User Scenarios

For URL analysis, the system downloads the submitted page, extracts the principal article text, and reuses the text-preprocessing, embedding, semantic-search, and classification components. Implementation-specific URL scoring details are omitted from the simplified pipeline figure for clarity. For community verification, authenticated users create threads, contribute evidence, comment, and vote; the resulting community verdict is stored separately from the machine-learning label. RSS ingestion is triggered six times per day by Celery Beat, after which new articles are collected, deduplicated, embedded, categorized, and stored.

TABLE I
PRINCIPAL FUNCTIONAL REQUIREMENTS

Code	Requirement	Priority
FR-01	Text, URL, and uploaded-image analysis	High
FR-02	Registration, JWT authentication, and Google OAuth	High
FR-03	Results with linguistic indicators and Gemini evidence	High
FR-04	Forum threads, comments, and voting	Medium
FR-05	Administrative user, content, and security management	High
FR-06	RSS-based automated news ingestion	High
FR-07	Password reset and email verification	High
FR-08	Real-time updates through WebSocket	Medium
FR-09	pHash, EXIF metadata extraction, and Gemini image analysis	Medium
FR-10	Feedback-based model-retraining workflow	Medium
FR-11	A/B testing for recommendation settings	Low
FR-12	User-interaction tracking and recommendations	Medium

TABLE II
PRINCIPAL NON-FUNCTIONAL REQUIREMENTS

Code	Category	Criterion
NFR-01	Security	JWT authentication, bcrypt hashing, Redis-based rate limiting, and audit logging
NFR-02	Performance	Asynchronous API and Celery processing to avoid blocking long-running requests
NFR-03	Deployment	Twelve production services defined in Docker Compose
NFR-04	Reliability	Isolated embedding service, safe UNKNOWN fallback, and atomic model replacement
NFR-05	Usability	Responsive React interface for desktop, tablet, and mobile devices
NFR-06	Privacy	IP addresses stored as SHA-256 hashes; user-agent data are not retained in audit records

III. ALGORITHM AND MACHINE LEARNING METHODOLOGY

The analysis architecture combines real-time semantic retrieval with asynchronous machine-learning analysis. Every symbol and abbreviated subscript used in the equations is defined in Table III before its first use.

A. Two-Stage Analysis Pipeline

1) *Stage 1: Real-Time Semantic Search*: The input text is converted into a 768-dimensional embedding and compared with embeddings in the labeled articles store through pgvector cosine-distance search. Only records whose stored status maps to FAKE or AUTHENTIC are eligible for Stage 1 voting. The threshold is set to 0.08, corresponding to a cosine similarity above 0.92. Up to three qualifying records participate in similarity-squared weighted voting. For the i th retrieved record,

$$s_i = 1 - d_i, \quad (1)$$

$$w_i = s_i^2. \quad (2)$$

The winning Stage 1 label is

$$\hat{y}_{S1} = \arg \max_{c \in \mathcal{C}} \sum_{i: y_i = c} w_i. \quad (3)$$

TABLE III
SYMBOLS AND ABBREVIATED SUBSCRIPTS USED IN THE EQUATIONS

Symbol	Definition	Symbol	Definition
i	Index of a retrieved record	c	Candidate class in $\mathcal{C}$
d_i	Cosine distance of the i th retrieved record	s_i	Cosine similarity, $s_i = 1 - d_i$
w_i	Similarity-squared voting weight	y_i	Label of the i th retrieved record
$\mathcal{C}$	Class set {FAKE, AUTHENTIC}	$\hat{y}_{S1}$	Winning label produced by Stage 1
x_{cb}	Clickbait score (cb : clickbait)	x_{ex}	Exclamation-mark ratio (ex : exclamation)
x_{uc}	Uppercase ratio (uc : uppercase)	x_{hd}	Hedging ratio (hd : hedging)
x_{qd}	Question-mark density (qd : question density)	x_{nd}	Numeric density (nd : numeric density)
x_{sw}	Short-word penalty (sw : short word)	x_{src}	Official-source score (src : source)
$\bar{\ell}$	Average word length	r_{raw}	Unclipped rule-based risk score
r	Rule-based risk score clipped to $[0, 1]$	p_f	Classifier probability assigned to FAKE
α	Classifier weight in the general local policy	z_α	Combined local decision score for weight α
z	Selected local decision score; $z = p_f$ when $\alpha = 1$	τ	Binary local decision threshold (0.50)
q	Operational confidence score, $\max(z, 1 - z)$	γ	Gemini prompt-mode threshold (0.65)
$\hat{y}_{ML}$	Local Stage 2 machine-learning prediction	$S1, ML$	Subscripts denoting Stage 1 and machine learning, respectively

TABLE IV
HANDCRAFTED NLP SIGNALS AND RISK-SCORE COEFFICIENTS

Signal	Definition	Coeff.
Clickbait	Sensational or clickbait expressions	+0.30
Exclamation	Exclamation-mark density	+0.20
Uppercase	Uppercase-character or word ratio	+0.15
Hedging	Uncertainty and hedging expressions	+0.15
Question	Question-mark density	+0.10
Numeric	Numerical-expression density	+0.05
Short-word	Penalty derived from average word length	+0.10
Source	References to recognized or official sources	-0.15

Squared-similarity weighting emphasizes stronger semantic matches while aggregating evidence from up to three records. The system stores the rank, similarity, cosine distance, normalized label, voting weight, and winner indicator for the participating records. The Stage 1 response reports the winning label, vote agreement, best-match similarity, and supporting record.

2) Stage 2: Asynchronous Machine-Learning Analysis:

If no record satisfies the Stage 1 threshold, a Celery task is queued. The worker preprocesses the text, regenerates the 768-dimensional embedding, extracts eight linguistic signals, and constructs a 776-dimensional feature vector. A StandardScaler-Logistic Regression pipeline produces p_f . The local result and its model, risk, policy, and decision-path metadata are stored before Gemini processing. For authenticated submissions, a local completion event is also published through WebSocket. If the classifier is unavailable or prediction fails, the system returns the operational status UNKNOWN rather than forcing a rule-based binary decision.

B. Natural Language Processing Signal Extraction

The eight handcrafted signals are transferred to the classifier in a fixed order through the `SIGNAL_KEYS` constant. Their risk-score coefficients are summarized in Table IV.

The short-word penalty is

$$x_{sw} = \max\left(0, \frac{5.5 - \bar{\ell}}{5.5}\right). \quad (4)$$

The raw risk score is

$$r_{raw} = 0.30x_{cb} + 0.20x_{ex} + 0.15x_{uc} + 0.15x_{hd} + 0.10x_{qd} + 0.05x_{nd} + 0.10x_{sw} - 0.15x_{src}. \quad (5)$$

The score is clipped to the unit interval:

$$r = \min(1, \max(0, r_{raw})). \quad (6)$$

C. Decision Policy

The general decision-policy equation evaluated in the ablation study is

$$z_\alpha = \alpha p_f + (1 - \alpha)r, \quad 0 \leq \alpha \leq 1. \quad (7)$$

The ablation results selected $\alpha = 1.0$ with the historical rule-based override disabled. Therefore, the current production policy is

$$z = z_{1.0} = p_f. \quad (8)$$

The binary local prediction and confidence score are

$$\hat{y}_{ML} = \begin{cases} \text{FAKE}, & z > \tau, \\ \text{AUTHENTIC}, & z \leq \tau, \end{cases} \quad \tau = 0.50, \quad (9)$$

$$q = \max(z, 1 - z). \quad (10)$$

The score q is an operational decision score and has not been post-hoc calibrated as a probability-confidence estimate. The implementation sets the Gemini decision flag when $q \leq \gamma$, where $\gamma = 0.65$. For $q > \gamma$, the prompt presents the local verdict as a higher-confidence result to be checked and explained. In both prompt modes, Gemini returns one of FAKE, AUTHENTIC, or IDDDIA; IDDDIA is mapped to the stored status UNCERTAIN. If the mapped Gemini verdict differs from the local verdict, the database status is updated asynchronously. Consequently, $\hat{y}_{ML}$ and q describe the local Stage 2 decision only and may differ from the result eventually displayed after Gemini processing.

TABLE V
DATASET AND EXPERIMENTAL CONFIGURATION

Parameter	Value
Raw sources	AA (197, AUTHENTIC), HT (1,397, AUTHENTIC), and Teyit (2,181, mixed labels)
Raw records	3,775
Evaluated samples	3,764 after preprocessing
Class distribution	1,759 AUTHENTIC; 2,005 FAKE
Protocol	Five-fold stratified cross-validation
Fold assignment	Identical folds; <code>random_state=42</code>
Classifier	Logistic Regression; balanced class weights; $C = 1.0$; 1,000 iterations
Preprocessing	StandardScaler fitted within each fold
Hybrid features	768-dimensional embedding + 8 signals
Primary metric	Macro-F1
Additional metrics	Accuracy, per-class recall, and FP rate
Source grouping	Not applied in the reported stratified folds
Duplicate grouping	Exact and near-duplicate claim groups were not enforced across folds
Statistical testing	No paired significance test was applied

D. Experimental Setup

All baseline, feature-ablation, and decision-policy experiments used the same fixed CSV sources and identical fold assignments. The files contained 3,775 raw rows; preprocessing removed records whose text became empty, leaving 3,764 evaluated samples. Table V summarizes the configuration.

A separate reproducible 80/20 holdout experiment on the same fixed dataset used 3,011 training and 753 test samples and yielded 87.52% accuracy. It is retained only to reproduce the previously reported rounded 88% value; five-fold cross-validation is the primary evaluation.

The reported baseline, feature-ablation, and decision-policy results evaluate the local Stage 2 classifier and its rule-based decision policy. Stage 1 direct matches, Gemini’s asynchronous review, URL-specific processing, and the image-analysis path are evaluated as separate system components. The fixed CSV snapshots are distinct from the live production database and from the classifier that may be replaced by the scheduled retraining workflow. Candidate values of α were compared using identical fold assignments to enable direct policy comparison.

IV. EXPERIMENTAL RESULTS

A. Baseline and Feature-Ablation Results

Table VI compares the majority baseline, TF-IDF, BERT-only, signal-only, and feature-level hybrid variants. All variants were evaluated on identical folds.

TF-IDF with Logistic Regression obtained the highest Macro-F1 score of 0.8745, while the feature-level hybrid model achieved 0.8724. The difference of 0.0021 indicates numerically close performance on the evaluated folds. Adding the eight handcrafted signals increased Macro-F1 from 0.8663 to 0.8724, corresponding to a gain of 0.0061 over BERT-only features.

The pooled out-of-fold confusion matrix contains 207 false-positive and 272 false-negative predictions. The false-positive rate of the selected local Stage 2 policy is 0.1177. These counts precede any asynchronous Gemini status update.

B. Decision-Policy Ablation

Table VIII summarizes selected values of α and the historical override setting.

The highest measured Macro-F1 was obtained with $\alpha = 1.0$ and the override disabled. This setting also increased FAKE recall while producing a higher false-positive rate than lower model weights. The selected local policy therefore favors stronger detection of fake items while retaining classifier probability as the sole local decision score. The historical override did not improve FAKE recall at $\alpha = 1.0$ and slightly reduced Macro-F1; it is not called by the current analysis worker.

C. Dataset Expansion Analysis

A dataset-expansion experiment added 483 usable records from the TeyitUltra archive, resulting in 4,247 evaluated samples. Five-fold cross-validation on the expanded dataset yielded an accuracy of 0.8886 and a Macro-F1 score of 0.8862. This analysis is reported separately from the fixed 3,764-sample baseline because it evaluates a different dataset composition.

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Microservice Architecture

The production deployment is organized into 12 Docker Compose services. An additional frontend container is defined for local development, whereas the production React frontend is deployed through Vercel. Table IX summarizes the production services.

The embedding service is isolated to separate model-memory consumption from the API and analysis-worker processes. This design reduces resource contention and permits model-specific memory limits without coupling BERT loading to every application process.

B. Database Design

PostgreSQL 15 was selected because pgvector allows 768-dimensional embeddings to be stored with relational data and searched using the `<=>` cosine-distance operator. This avoids a separate vector-database service and allows relational and vector queries to share the same connection pool. Figure 2 presents selected entities and relationships; secondary tables and implementation-level constraints are omitted for readability.

1) *User, Analysis, and Content Tables*: The `users` table stores account, authentication, trust-tier, XP, and level information. Each `analysis_request` may reference at most one result through a nullable `result_id`. A previously generated result can be referenced by multiple requests because this foreign key is not unique. The request row stores the user reference, anonymized IP hash, analysis type, task identifier, result reference, and creation time; it does not duplicate raw submitted content.

Each `analysis_result` references exactly one article. The corresponding `article_id` is unique, so an article has at most one analysis-result row. Results store the operational

TABLE VI
BASELINE AND FEATURE-ABLATION RESULTS ON THE FIXED 3,764-SAMPLE DATASET (MEAN $\pm$ STANDARD DEVIATION ACROSS FIVE FOLDS)

Method	Accuracy	Macro-F1	Recall (FAKE)	Recall (AUTH.)
Majority-class baseline	0.5327 $\pm$ 0.0003	0.3475 $\pm$ 0.0001	1.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
TF-IDF + Logistic Regression	0.8757 $\pm$ 0.0093	0.8745 $\pm$ 0.0095	0.9122 $\pm$ 0.0101	0.8340 $\pm$ 0.0177
BERT embeddings only (768-dim)	0.8666 $\pm$ 0.0063	0.8663 $\pm$ 0.0062	0.8569 $\pm$ 0.0164	0.8778 $\pm$ 0.0113
NLP signals only (8-dim)	0.6918 $\pm$ 0.0181	0.6907 $\pm$ 0.0186	0.5955 $\pm$ 0.0305	0.8016 $\pm$ 0.0099
BERT + NLP signals (776-dim)	0.8727 $\pm$ 0.0065	0.8724 $\pm$ 0.0065	0.8643 $\pm$ 0.0183	0.8823 $\pm$ 0.0138

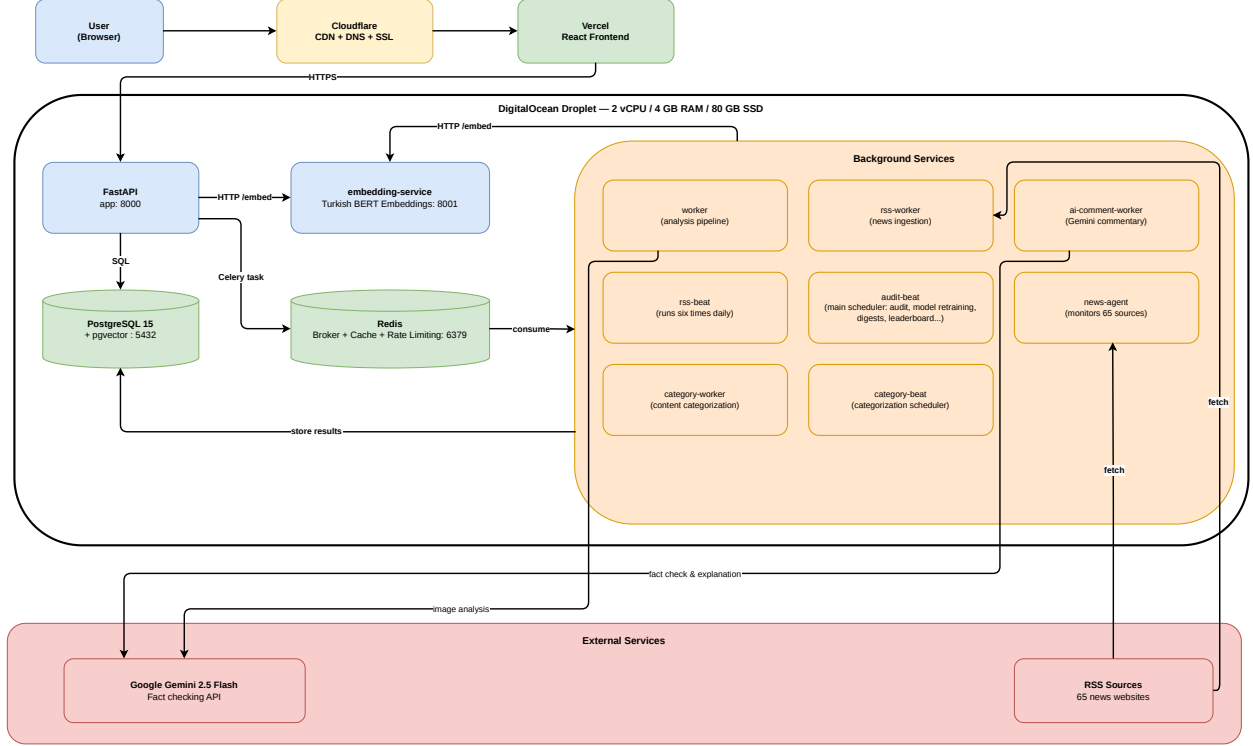


Fig. 1. System architecture block diagram.

TABLE VII
POOLED OUT-OF-FOLD CONFUSION MATRIX FOR THE 776-DIMENSIONAL HYBRID MODEL

	Pred. AUTH.	Pred. FAKE
Actual AUTHENTIC	1552	207
Actual FAKE	272	1733

status, confidence, model probability, rule-based risk score, combined score, model and policy versions, decision path, linguistic signals, and optional Gemini or full-report outputs. The status is initially written from the local decision and can later be replaced by the mapped Gemini verdict; the separately stored model probability, risk score, policy version, and decision path preserve the local-decision metadata. Operational statuses can include FAKE, AUTHENTIC, UNKNOWN, and UNCERTAIN.

The `analysis_matches` table stores the records participating in Stage 1 voting. For each request, the application records up to three matches with their rank, similarity, cosine distance, normalized label, voting weight, and winner indicator. The `articles` table forms the labeled semantic-search store and contains article content, source metadata, statuses, and 768-dimensional embeddings. Stage 1 excludes records whose status does not map to a binary FAKE/AUTHENTIC label. Live store size is reported separately from the fixed experimental dataset because the production database can grow over time.

2) *Image, Forum, and Audit Tables*: The `image_cache` table stores perceptual hashes, EXIF metadata, and Gemini visual-analysis output. The image pipeline first checks the pHash cache. If no reusable match exists, EXIF metadata are extracted and retained as contextual information; EXIF does not independently determine the verdict. Gemini 2.5 Flash then

TABLE VIII
SELECTED LOCAL DECISION-POLICY ABLATION RESULTS; THE CURRENT STAGE 2 POLICY IS HIGHLIGHTED IN BOLD

α	Override	Accuracy	Macro-F1	Recall (FAKE)	Recall (AUTH.)	FP Rate
0.55	Off	0.8409	0.8408	0.7751	0.9159	0.0841
0.55	On	0.8401	0.8400	0.7751	0.9142	0.0858
0.70	Off	0.8624	0.8623	0.8319	0.8971	0.1029
0.80	Off	0.8666	0.8664	0.8484	0.8874	0.1126
0.90	Off	0.8711	0.8709	0.8589	0.8852	0.1148
1.00	Off	0.8727	0.8724	0.8643	0.8823	0.1177
1.00	On	0.8722	0.8719	0.8643	0.8812	0.1188

TABLE IX
PRODUCTION DOCKER SERVICES

Service	Port	Type	Primary Role
embedding-service	8001	Model service	Turkish BERT embedding endpoint
app	8000	API service	FastAPI routes, authentication, and task submission
worker	–	Celery worker	Text analysis and deep-report tasks
ai-comment-worker	–	Celery worker	Gemini evidence and commentary tasks
rss-worker	–	Celery worker	RSS ingestion tasks
category-worker	–	Celery worker	Content-categorization tasks
rss-beat	–	Scheduler	Periodic RSS-task publication
category-beat	–	Scheduler	Periodic categorization tasks
audit-beat	–	Scheduler	Audit, retraining, digest, and maintenance scheduling
news-agent	–	Background agent	Monitoring of 65 news sources
db	5432	Database	PostgreSQL 15, pgvector, and persistent data
redis	6379	Broker/cache	Celery broker, cache, rate limiting, and transient events

performs visual analysis, and the result is cached.

Forum tables store threads, comments, votes, tags, reports, and moderation state. Nested comments and community verdicts are maintained independently of machine-learning predictions. Audit events are first transferred to a Redis buffer and later flushed to PostgreSQL by a scheduler. Moving database writes outside the synchronous request path is intended to reduce request-processing overhead; no latency claim is made without a dedicated benchmark.

C. Analysis Pipeline Implementation

Figure 3 presents a simplified view of the analysis pipeline.

1) *Synchronous Stage*: Directly entered text and text extracted from a URL are preprocessed and represented using a 768-dimensional embedding. The embedding is compared with the labeled `articles` store. If qualifying records are found, similarity-squared weighted voting returns the Stage 1 label and supporting match information. Uploaded images are routed to a separate pHash-cache and Gemini visual-analysis path. URL-specific scoring and EXIF implementation details are omitted from the figure for clarity.

2) *Asynchronous Stage*: If Stage 1 does not find a qualifying match, the text is submitted to a Celery worker. The worker regenerates the embedding, extracts the eight signals, constructs the 776-dimensional vector, and produces the local classifier probability. The local result is stored and, for authenticated submissions, a completion event is delivered through WebSocket. Gemini is subsequently invoked with Google Search grounding. For $q \leq 0.65$, the prompt asks Gemini to decide a locally uncertain case; otherwise, it asks Gemini to verify and explain the local result. In either mode,

a differing Gemini verdict can replace the stored status, while the evidence and commentary are added asynchronously.

D. API and Community Components

The REST API is organized into authentication, analysis, forum, user-management, administration, gamification, news, messaging, notification, sharing, reporting, recommendation, A/B-testing, source, and infrastructure route groups. Most user-specific and administrative operations require JWT authentication, while authentication, health-check, and selected public-sharing routes remain publicly accessible. Route and service counts are generated automatically from the application and Docker configuration to prevent documentation drift.

The platform includes XP, levels, badges, trust tiers, community voting, direct messaging, and leaderboards. These components support user participation but are not used as input features for the classifier evaluated in this paper. Community-voting labels and machine-learning labels remain separate taxonomies.

E. User Interface

The React and Tailwind CSS interface provides responsive analysis, result, report, forum, profile, news, and administrative pages. Figure 4 shows the principal result and report views.

VI. LIMITATIONS AND FUTURE WORK

The present evaluation is based on a fixed dataset collected from a limited set of Turkish-language sources. Accordingly, the reported cross-validation results primarily characterize performance within this dataset composition. The experiments focus on the local Stage 2 classifier, while Stage 1 semantic

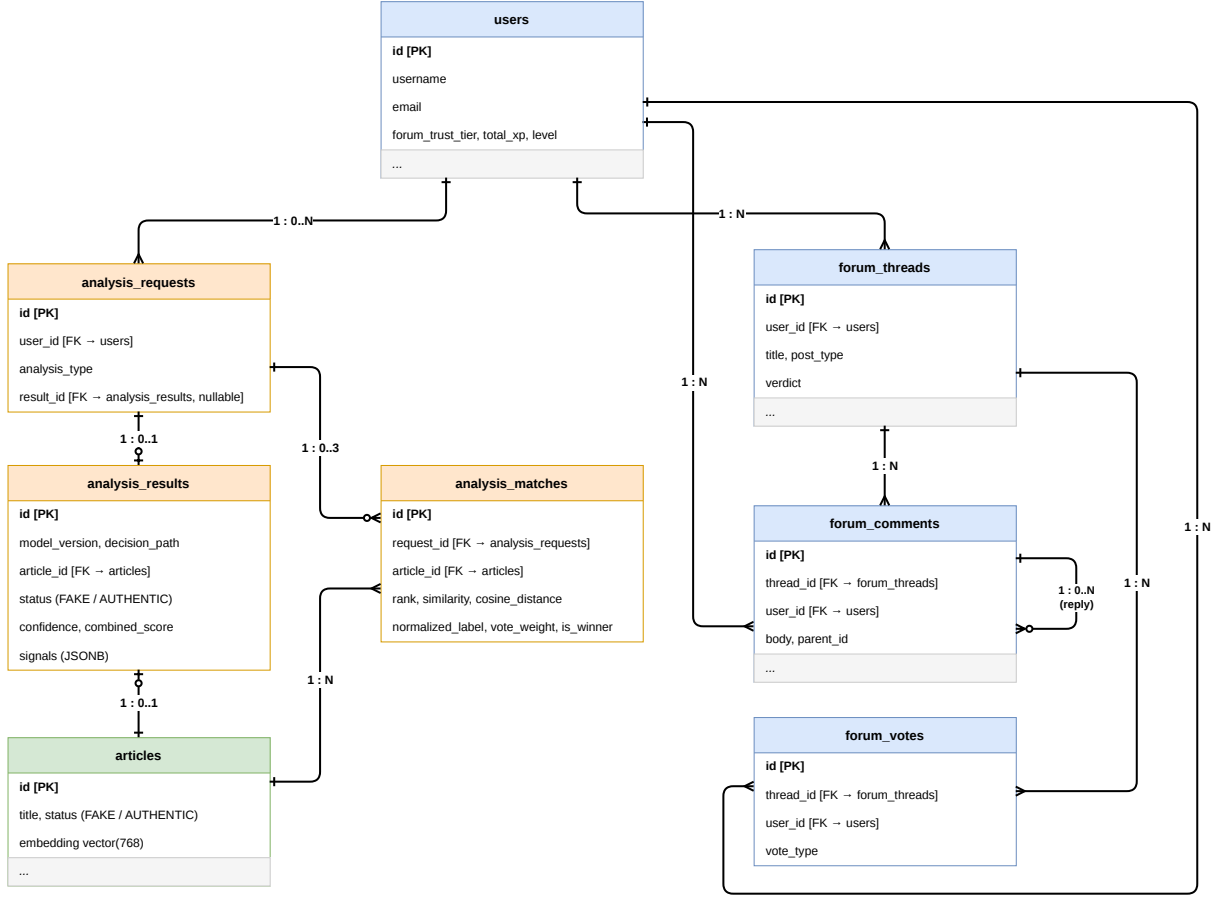


Fig. 2. Simplified logical entity-relationship diagram of the principal database tables.

retrieval, Gemini-assisted review, URL analysis, and image analysis are treated as separate system components. In addition, the binary FAKE/AUTHENTIC formulation provides an operational simplification of the broader information-disorder taxonomy, and the local confidence score is used as an operational score rather than a calibrated probability estimate. Gemini-generated outputs may also vary depending on the configured model version and the web evidence available at execution time.

Future work will extend the evaluation through source-aware and temporal test settings, duplicate-aware data partitioning, and an independent external test set. Further studies will examine Stage 1 retrieval thresholds, the contribution of Gemini-assisted review, confidence calibration, and end-to-end system performance. Latency and load measurements will also be conducted to characterize deployment performance. The final tagged release will document the experimental data configuration, software dependencies, model revision, and machine-readable result files to support reproducibility.

VII. CONCLUSION

This study presented nehaber.dev, an integrated Turkish-language platform combining semantic retrieval, feature-level

machine-learning classification, LLM-assisted evidence generation, URL and uploaded-image analysis, asynchronous processing, and community verification. On the fixed 3,764-sample dataset, the local 776-dimensional Stage 2 classifier achieved 0.8727 ± 0.0065 accuracy and 0.8724 ± 0.0065 Macro-F1 under five-fold stratified cross-validation. The eight handcrafted signals produced a numerical Macro-F1 gain of 0.0061 over BERT-only features. TF-IDF with Logistic Regression obtained a slightly higher Macro-F1 of 0.8745 ± 0.0095 , showing that the classical lexical baseline remained highly competitive on the evaluated dataset.

Decision-policy ablation selected classifier probability alone ($\alpha = 1.0$) with the historical override disabled for the local Stage 2 decision. This configuration achieved the highest measured Macro-F1 and FAKE recall among the tested local policies, although it also produced a higher false-positive rate than lower model weights. The reported values do not measure the complete production outcome because Stage 1 direct matches and subsequent Gemini status updates were excluded. The primary contribution is therefore not a claim of source-independent superiority over classical baselines, but the implementation and evaluation of an operational platform that

integrates semantic retrieval, local classification, asynchronous evidence review, and community participation for Turkish content.

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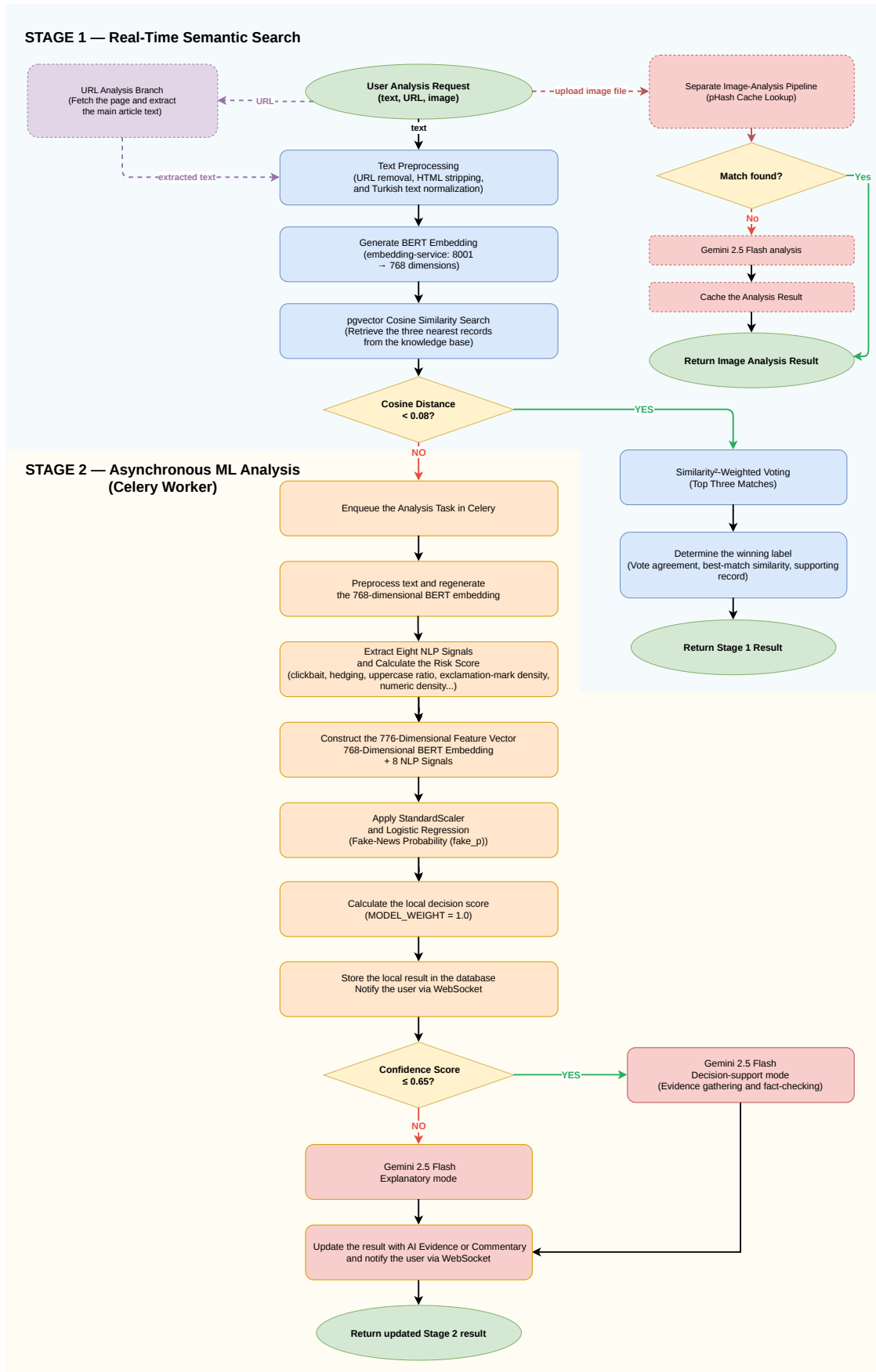
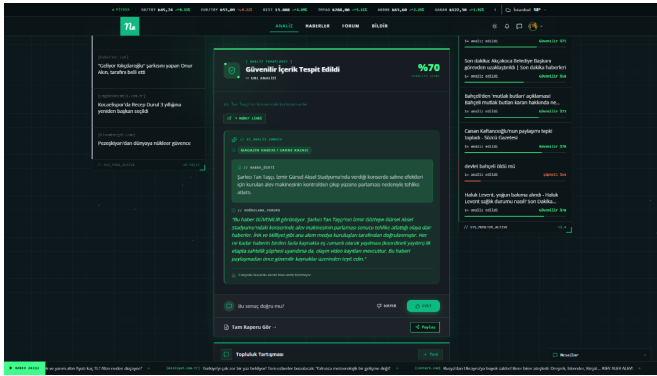
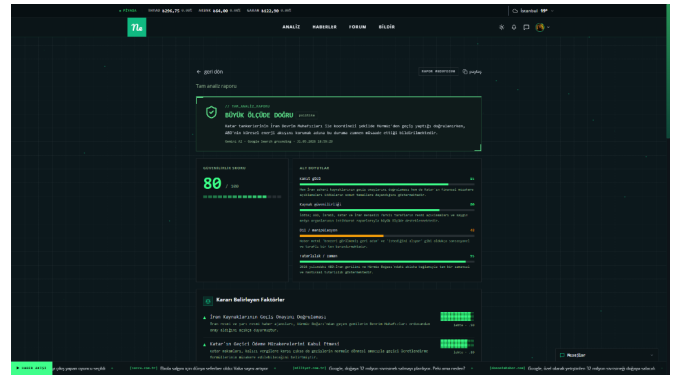


Fig. 3. Simplified two-stage analysis pipeline. URL-specific scoring and EXIF metadata handling are omitted for clarity.



(a) Analysis result



(b) Detailed analysis report

Fig. 4. Representative user-interface views.